Lab 2 FDSAI

2026-2027

1. Draw n = 1000 random observations $x_1, \ldots, x_n$ from a distribution you like. Repeat 500 and 1000 times.

   For each of the 1000 samples, compute the sample mean $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ and fill the following table:

	$x_1 \; x_2 \; \ldots x_n$	$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$
t=1		
…		
t=1000		

   Plot a histogram for the 1000 sample means thus obtained. What do you observe?

2. Prepare the dataset (https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBMDeveloperSkillsNetwork-ST0151EN-SkillsNetwork/labs/teachingratings.csv) for a statistical study:
   a) Provide an overview of the characteristics present in the dataset. Specify the type of each variable.
   b) Encode the categorical variables if their values are not numeric.
   c) Identify and handle (if any) the outliers in the dataset. Choose whichever method you prefer but provide a justification for your choice.
3. Based on the previous dataset, create a plot that meets the following criteria:

- On the **x-axis**: the first 10 values of the variable **prof**.
- On the **y-axis**: the **average rating** obtained by each professor across all their subjects.
  *(Hint: the column **eval** contains the professor's rating for a specific subject.)*